TokoBli Campaign Evaluation & Product Page Analysis

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08 November 2025
RevoU FSDA batch OC2025

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Business Overview & Disclaimer

Company Overview:
TokoBli is Indonesia's leading e-commerce platform, offering diverse product categories and running major sales campaigns on "twin dates". The business aims to leverage data analytics to optimize campaign strategies, grow transactions, and maximize revenue while keeping campaign costs efficient.

Disclaimer:
Analysis performed on provided e-commerce dataset for educational demonstration.
Insights and recommendations are based strictly on TokoBli dataset.

Dataset Overview:
Main tables and essential fields.

Project Goal:

1. Evaluate the previous three campaign scenarios and provide actionable insights.
2. Experiment whether the new product detail page will bring more transactions than the current product detail page.

1. Business Understanding

A. What is the core business problem faced by TokoBli?

Bussiness Problem:

TokoBli is a leading e-commerce platform in Indonesia, offering a wide range of product categories and running massive sales campaigns on two dates in the last three months with different strategies, resulting in 10x more transactions.

Based on the existing business problem, we were able to analyze TokoBli's dataset to select the most optimal campaign strategy from three campaigns the company ran on ‘twin dates’ (10/10, 11/11, 12/12).

1. Business Understanding

A. What is the core business problem faced by TokoBli?

Business Question:

How can we determine the most effective campaign strategies in terms of revenue, customer engagement, and discount cost efficiency, and how can we use these insights to optimize future promotional campaigns?

Problem Statement

We aim to evaluate the performance of the last three campaign strategies by comparing key business metrics such as total transactions, number of customers, products sold, total revenue, and discount budget usage. campaign budget for discounts, transaction count, customer count, products sold, and total revenue. The goal is to identify which campaign yields the highest revenue efficiency within the next reporting cycle, so that future campaign planning can be more data-driven and cost-effective.

1. Business Understanding

B. Considering the data that you have, how do you think the data can help you answer the business problem?

The dataset provides key performance indicators that allow us to compare the effectiveness of each campaign strategy:

- **Total Transactions per Campaign:** We understand how many purchases were made and how attractive each campaign was in driving customer activity.
- **Customer Count:** Shows how many unique customers participated in each campaign, allowing us to evaluate audience reach and campaign engagement.
- **Products Sold:** Indicates product movement and demand level during each campaign, helping us identify which promotional approach drives higher product turnover.
- **Total Revenue:** Allows us to measure the monetary impact of each campaign and compare financial outcomes across strategies.
- **Discount Budget / Promotion Cost:** Helps us determine how much cost was spent to achieve the campaign results and calculate revenue efficiency (return relative to promo budget).

1. Business Understanding

C. What is the statistics that you think is best to answer business problems?

Back to our main objective to determine the most effective campaign strategy, the descriptive statistics we can propose are:

We can calculate the average Total Revenue, average Quantity sold, and the average Discount usage for each campaign. To make sure the comparison is fair, we should also examine data distribution and detect potential outliers using Quartile analysis. This will allow us to understand the central tendency of campaign performance while minimizing bias from extreme values.

By comparing these averages across the three campaigns, we can identify which campaign generated higher revenue and product movement with efficient discount spending, giving us a clearer insight into the most effective promotional strategy.

2. Data Cleaning

A. Handle missing data in column

Salinan dari [STUDENT] TokoBli E-commerce Data

File Edit Tampilan Sisipkan Format Data Alat Ekstensi Bantuan Minta Gemini

A1:N1 | Transaction ID

Transaction	Product ID	Customer ID	Status	Created At	Campaign	SKU	Price	QTY	Discount	Shipping Cost	Total Revenue	Category Name	Payment Meth
ID-1	724241	21	complete	2023-10-10	Campaign 10/10	BAGLUX5A099F4E3D5CF	2.10E+06	5	0.00E+00	0	10,500,000	Beauty & Grooming	COD
ID-2	724074	168	complete	2023-10-10	Campaign 10/10	SOGAMB5A311E53D6BE9	4.20E+06	1	0.00E+00	0	4,200,000	Soghaat	COD
ID-3	724075	168	complete	2023-10-10	Campaign 10/10	SOGAMB5A09BA2372A3B	1.20E+06	1	0.00E+00	0	1,200,000	Soghaat	COD
ID-4	724076	168	complete	2023-10-10	Campaign 10/10	BAGGAR5A0A925948525	3.97E+06	1	0.00E+00	0	3,970,000	Beauty & Grooming	COD
ID-5	724077	168	complete	2023-10-10	Campaign 10/10	KNO59D64DAF36939	2.70E+05	2	0.00E+00	0	540,000	Superstore	COD
ID-6	724078	168	complete	2023-10-10	Campaign 10/10	PSO5A43482AF3C58	2.00E+05	1	0.00E+00	0	200,000	Superstore	COD
ID-7	724578	6715	complete	2023-10-10	Campaign 10/10	MEFTHE59F6D761A69C1-40	7.99E+06	1	0.00E+00	0	7,990,000	Men's Fashion	COD
ID-8	724442	8616	complete	2023-10-10	Campaign 10/10	WOFCHA5A267D9B0BF85	8.99E+06	1	0.00E+00	0	8,990,000	Women's Fashion	COD
ID-9	724443	8616	complete	2023-10-10	Campaign 10/10	WOFDEA59CE228149430-36	4.49E+06	1	0.00E+00	0	4,490,000	Women's Fashion	COD
ID-10	724062	8765	complete	2023-10-10	Campaign 10/10	BAGDOV5A099F4EF3FFA	3.70E+06	1	0.00E+00	0	3,700,000	Beauty & Grooming	COD
ID-11	724896	15960	complete	2023-10-10	Campaign 10/10	SOGAMB5A2E671470128	9.00E+05	2	0.00E+00	0	1,800,000	Soghaat	COD
ID-12	724897	15960	complete	2023-10-10	Campaign 10/10	SOGAMB5A2E67122360F	9.00E+05	2	0.00E+00	0	1,800,000	Soghaat	COD
ID-13	724898	15960	complete	2023-10-10	Campaign 10/10	SOGAMB5A2E6713BB9AD	9.00E+05	2	0.00E+00	0	1,800,000	Soghaat	COD
ID-14	724899	15960	complete	2023-10-10	Campaign 10/10	SOGMAY5A2F7FC4D8DCA	2.20E+06	1	0.00E+00	0	2,200,000	Soghaat	COD
ID-15	724900	15960	complete	2023-10-10	Campaign 10/10	SOGMAY5A2F7FD653351	4.80E+06	1	0.00E+00	0	4,800,000	Soghaat	COD
ID-16	724901	15960	complete	2023-10-10	Campaign 10/10	SOGSND5A43754748823	9.00E+05	1	0.00E+00	0	900,000	Soghaat	COD
ID-17	724902	15960	complete	2023-10-10	Campaign 10/10	SOGSND5A4375431C7D4	1.70E+06	1	0.00E+00	0	1,700,000	Soghaat	COD
ID-18	724903	15960	complete	2023-10-10	Campaign 10/10	SOGSND5A437540871D2	2.20E+06	1	0.00E+00	0	2,200,000	Soghaat	COD
ID-19	724904	15960	complete	2023-10-10	Campaign 10/10	SOGAMB5A2E6716F1D40	9.00E+05	2	0.00E+00	0	1,800,000	Soghaat	COD
ID-20	724905	15960	complete	2023-10-10	Campaign 10/10	SOGAMB5A2E671507BC4	9.00E+05	1	0.00E+00	0	900,000	Soghaat	COD
ID-21	724738	16593	complete	2023-10-10	Campaign 10/10	BAGLUX5A099F4E3D5CF	2.10E+06	1	2.10E+05	0	1,890,000	Beauty & Grooming	COD
ID-22	724266	17684	complete	2023-10-10	Campaign 10/10	MEFTHE59F6D780CDF16-43	5.99E+06	1	5.99E+05	0	5,391,000	Men's Fashion	COD
ID-23	724268	17684	complete	2023-10-10	Campaign 10/10	BAGKEM5A38AE901D18E	4.45E+06	1	4.45E+05	0	4,005,000	Beauty & Grooming	COD
ID-24	724665	18967	complete	2023-10-10	Campaign 10/10	MEFDEA59F1EE0AD3865-M	5.48E+06	1	0.00E+00	0	5,480,000	Men's Fashion	COD
ID-25	724667	18967	complete	2023-10-10	Campaign 10/10	MEFDEA5A44C0264285B-M	3.99E+06	1	0.00E+00	0	3,990,000	Men's Fashion	COD
ID-26	724669	18967	complete	2023-10-10	Campaign 10/10	MEFDEA59CA5159DB6EB-M	1.20E+06	1	0.00E+00	0	1,200,000	Men's Fashion	COD
ID-27	724073	19100	complete	2023-10-10	Campaign 10/10	ENTYUE59FABFA53B9AA	5.49E+06	1	0.00E+00	0	5,490,000	Entertainment	COD
ID-28	724513	19168	complete	2023-10-10	Campaign 10/10	BAGEMA59BAB1182E004	8.00E+06	1	8.00E+05	0	7,200,000	Beauty & Grooming	Transfer Bank
ID-29	724516	19168	complete	2023-10-10	Campaign 10/10	HALEMA59BA8E048DC2C	3.00E+06	1	3.00E+05	0	2,700,000	Home & Living	Transfer Bank
ID-30	724517	19168	complete	2023-10-10	Campaign 10/10	APPANE59AC41CF5140F	6.74E+06	1	6.74E+05	0	6,066,000	Appliances	Transfer Bank

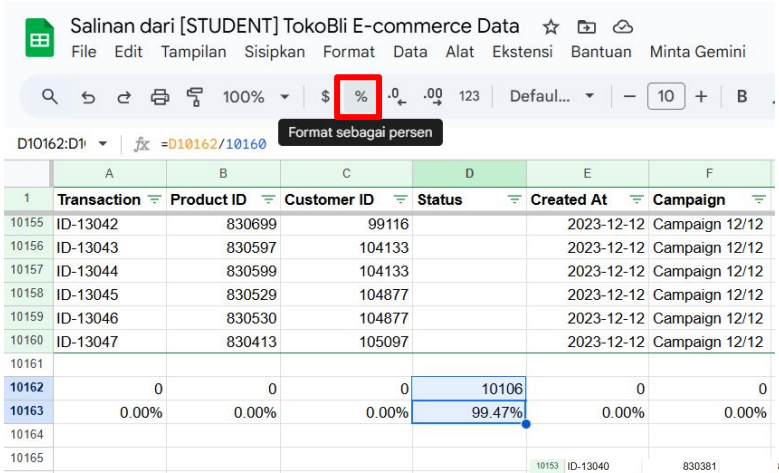
Dataset | Dataset Cleaning | Data Dictionary

Cell Berisi: 14

This is the raw data provided to answer our business problem Rows Count (with header) : 10.160 | Columns Count : 14

2. Data Cleaning

A. Handle missing data in rows



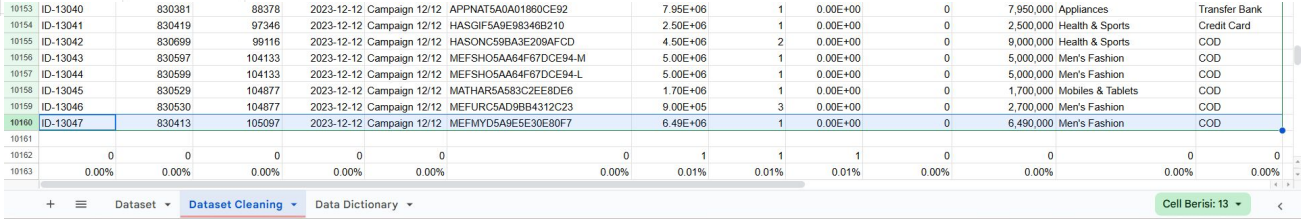
	A	B	C	D	E	F
1	Transaction	Product ID	Customer ID	Status	Created At	Campaign
10155	ID-13042	830699	99116		2023-12-12	Campaign 12/12
10156	ID-13043	830597	104133		2023-12-12	Campaign 12/12
10157	ID-13044	830599	104133		2023-12-12	Campaign 12/12
10158	ID-13045	830529	104877		2023-12-12	Campaign 12/12
10159	ID-13046	830530	104877		2023-12-12	Campaign 12/12
10160	ID-13047	830413	105097		2023-12-12	Campaign 12/12
10161						
10162	0	0	0	10106	0	0
10163	0.00%	0.00%	0.00%	99.47%	0.00%	0.00%
10164						
10165						

U can drag down the grey line, so that the name of row does not disappear when scrolled down

Go to the second row **after the last row** of column A >> type the **countblank function** >> select the cell range from A2:A10160. Under the countblank row, type =A10162/10160, then **click on % symbol** to get the presentation of it.

Block the rows of A10162 and A10163 then copy to all the column beside it. If theres **over 50% missing column** u can delete it.

countblank Is function to count the number of blank cells. A10162 is the place u type function of countblank.



10153	ID-13040	830381	88378	2023-12-12	Campaign 12/12	APPNAT5A0A01860CE92	7.95E+06	1	0.00E+00	0	7,950,000	Appliances	Transfer Bank
10154	ID-13041	830419	97346	2023-12-12	Campaign 12/12	HASGIF5A9E98346B210	2.50E+06	1	0.00E+00	0	2,500,000	Health & Sports	Credit Card
10155	ID-13042	830699	99116	2023-12-12	Campaign 12/12	HASONC59BA3E209AFCD	4.50E+06	2	0.00E+00	0	9,000,000	Health & Sports	COD
10156	ID-13043	830597	104133	2023-12-12	Campaign 12/12	MEFSHO5AA64F67DCE94-M	5.00E+06	1	0.00E+00	0	5,000,000	Men's Fashion	COD
10157	ID-13044	830599	104133	2023-12-12	Campaign 12/12	MEFSHO5AA64F67DCE94-L	5.00E+06	1	0.00E+00	0	5,000,000	Men's Fashion	COD
10158	ID-13045	830529	104877	2023-12-12	Campaign 12/12	MATHAR5A583C2EE8DE6	1.70E+06	1	0.00E+00	0	1,700,000	Mobiles & Tablets	COD
10159	ID-13046	830530	104877	2023-12-12	Campaign 12/12	MEFURC5AD9BB4312C23	9.00E+05	3	0.00E+00	0	2,700,000	Men's Fashion	COD
10160	ID-13047	830413	105097	2023-12-12	Campaign 12/12	MEFMYD5A9E5E30E80F7	6.49E+06	1	0.00E+00	0	6,490,000	Men's Fashion	COD
10161													
10162	0	0	0	0	0	0	1	1	1	0	0	0	0
10163	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.01%	0.01%	0.01%	0.00%	0.00%	0.00%	0.00%

Dataset | Dataset Cleaning | Data Dictionary | Cell Berisi: 13

Column after Handle missing data in rows: 13

2. Data Cleaning

A. Handle missing data in column

N1:O10160 | fx Blank

	J	K	L	M	N	O
1	Shipping Cost	Total Revenue	Category Name	Payment Meth	Blank	Blank%
2	0	10,500,000	Beauty & Grooming	COD	0	0.00%
3	0	4,200,000	Soghaat	COD	0	0.00%
4	0	1,200,000	Soghaat	COD	0	0.00%
5	0	3,970,000	Beauty & Grooming	COD	0	0.00%
6	0	540,000	Superstore	COD	0	0.00%
7	0	200,000	Superstore	COD	0	0.00%
8	0	7,990,000	Men's Fashion	COD	0	0.00%
9	0	8,990,000	Women's Fashion	COD	0	0.00%
10	0	4,490,000	Women's Fashion	COD	0	0.00%
11	0	3,700,000	Beauty & Grooming	COD	0	0.00%
12	0	1,800,000	Soghaat	COD	0	0.00%
13	0	1,800,000	Soghaat	COD	0	0.00%
14	0	1,800,000	Soghaat	COD	0	0.00%
15	0	2,200,000	Soghaat	COD	0	0.00%
16	0	4,800,000	Soghaat	COD	0	0.00%
17	0	900,000	Soghaat	COD	0	0.00%
18	0	1,700,000	Soghaat	COD	0	0.00%
19	0	2,200,000	Soghaat	COD	0	0.00%

Go to the second column **after the last column** column 2 >> type the countblank function >> select the cell range from A2:M2. Beside the countblank column, type =N2/13, then click on % symbol to get the presentation of it.

Block the colums of N2 and N3 then copy to all the rows under it >> Select All >> Click Filter >> On Blank% select option that more than 0% if want to see which column that blank.

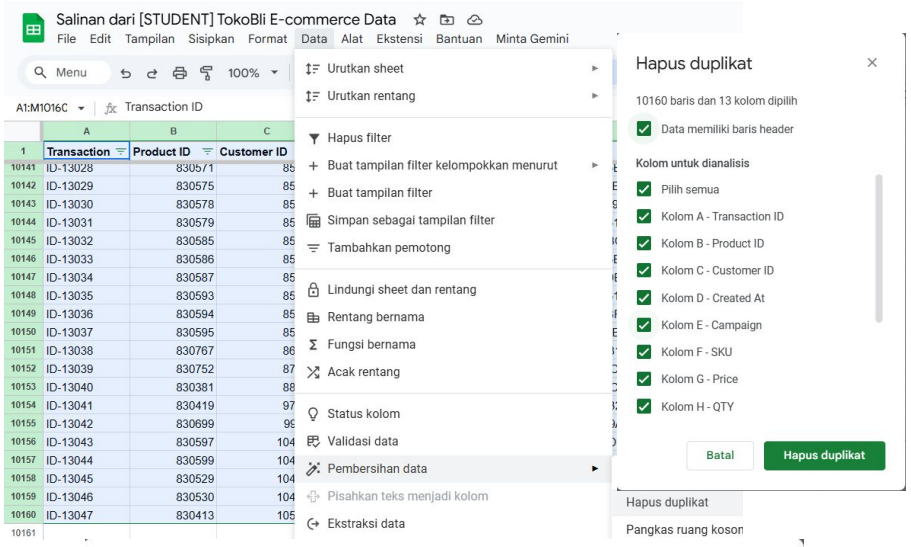
Decided to ignore the blank column

F	G	H	I	J	K	L	M	N	O
SKU	Price	QTY	Discount	Shipping Cost	Total Revenue	Category Name	Payment Meth	Blank	Blank%
SADPAR59F6C59E02C06		1	0.00E+00	0	2,500,000	School & Education	COD	1	7.69%
HALBUK5A013F87DB79F	1.76E+06		0.00E+00	0	1,760,000	Home & Living	Transfer Bank	1	7.69%
WOFDEA59CE2178B2F7E	1.99E+06	1		0	1,791,000	Women's Fashion	COD	1	7.69%
0	1	1	1	0	0	0	0		
0.00%	0.01%	0.01%	0.01%	0.00%	0.00%	0.00%	0.00%		

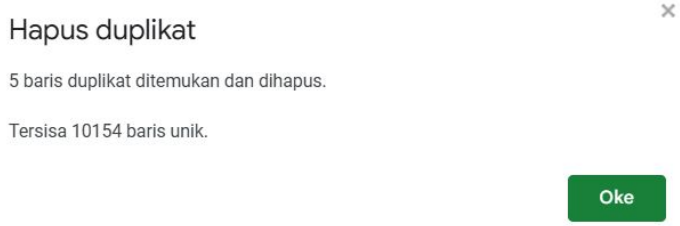
Column after Handle missing data in rows: 13

2. Data Cleaning

B. Remove Duplicated Data



Select all cells on tabs dataset cleaning >> click ribbon menu Data >> Data Cleanup >> Remove Duplicate >> checklist data has header row >> click Remove Duplicate.



Rows after remove duplicated data: 10.155 (with header)

2. Data Cleaning

C. Adjusting the data types of columns that are not suitable

Table1

	Transaction ID	Product ID	Customer ID	Created At	Campaign	SKU	Price	QTY	Discount	Shipping Cost	Total Revenue	Category Name	Payment M
6	ID-5	724077	168	10/10/2023	Campaign 10/10	KN059D64DAF36939	Rp270,000	2	Rp0	Rp0	Rp540,000	Superstore	COD
7	ID-6	724078	168	10/10/2023	Campaign 10/10	PSO5A43482AF3C58	Rp200,000	1	Rp0	Rp0	Rp200,000	Superstore	COD
8	ID-7	724578	6715	10/10/2023	Campaign 10/10	MEFTHE59F6D761A69C1-40	Rp7,990,000	1	Rp0	Rp0	Rp7,990,000	Men's Fashion	COD
9	ID-8	724442	8616	10/10/2023	Campaign 10/10	WOFCHA5A267D9B0BF85	Rp8,990,000	1	Rp0	Rp0	Rp8,990,000	Women's Fashion	COD
10	ID-9	724443	8616	10/10/2023	Campaign 10/10	WOFDEA59CE228149430-36	Rp4,490,000	1	Rp0	Rp0	Rp4,490,000	Women's Fashion	COD
11	ID-10	724062	8765	10/10/2023	Campaign 10/10	BAGDOV5A099F4EF3FFA	Rp3,700,000	1	Rp0	Rp0	Rp3,700,000	Beauty & Grooming	COD
12	ID-11	724896	15960	10/10/2023	Campaign 10/10	SOGAMB5A2E671470128	Rp900,000	2	Rp0	Rp0	Rp1,800,000	Soghaat	COD
13	ID-12	724897	15960	10/10/2023	Campaign 10/10	SOGAMB5A2E67122360F	Rp900,000	2	Rp0	Rp0	Rp1,800,000	Soghaat	COD
14	ID-13	724898	15960	10/10/2023	Campaign 10/10	SOGAMB5A2E6713BB9AD	Rp900,000	2	Rp0	Rp0	Rp1,800,000	Soghaat	COD
15	ID-14	724899	15960	10/10/2023	Campaign 10/10	SOGMAY5A2F7FC4D8DCA	Rp2,200,000	1	Rp0	Rp0	Rp2,200,000	Soghaat	COD

Reformat data type by clicking ribbon menu Format > Convert to Table > Turn Off Filter > Choose data type for each column.

To change the column format to suit, we need to look at the data type for each variable in the data dictionary tab.

For numeric data types in currency, select Column Format > Number > Currency. If your currency is Indonesian Rupiah, select Format > Number > Indonesian Rupiah. For category categorical I prefer choose format ‘text’. For date data types > select Column Format > Date > Date.

3. Descriptive Analytics

A. Calculate the lower bound and upper bound "Total Revenue" with the correct values & Eliminate all outliers data based on the column "Total Revenue"

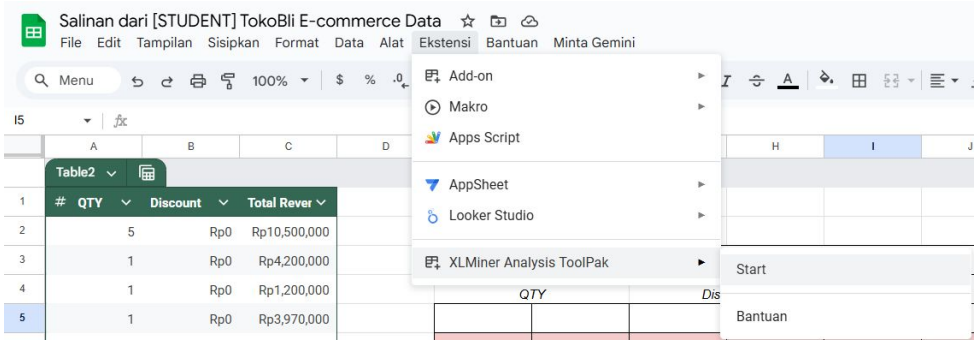
F4:G4 | fx =D4+(1.5*E4)

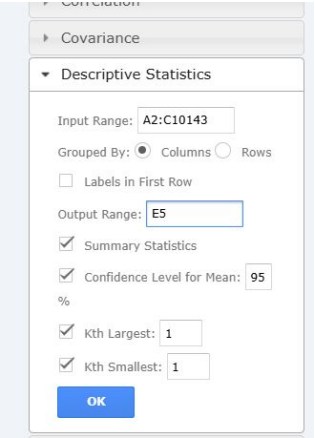
	A	B	C	D	E	F	G
	Outliers						
1	Total Rever	Is_Outliers					
2	Rp10,500,000	No					
3	Rp4,200,000	No	Q1	Q3	IQR	Upper Bound	Lower Bound
4	Rp1,200,000	No	2241000	6987500	4746500	14107250	-4878750
5	Rp3,970,000	No					0
6	Rp540,000	No					
7	Rp200,000	No					

Take the Total Revenue column from Dataset Cleaning > Calculate Q1 > Calculate Q3 > Calculate IQR > Determine the bounds > Lower Bound = Q1 − 1.5 × IQR > Upper Bound = Q3 + 1.5 × IQR. Remove or Delete Rows **above 14107250** But negative value in NumofPurchase is irrational, therefore we treat the LIF as 0.

3. Descriptive Analytics

B. Create Statistical measurement on the important column





To create a descriptive analysis, take the important columns needed and then move them to a new tab > click the extension menu ribbon > start.

when XLMiner Extension is open > select descriptive analysis tab > select data range > click input box > select new column for result > click output box > OK

Note: Value of Total Revenue obtained from the results of removing outliers that have been done

3. Descriptive Analytics

B. Interpret Statistical measurement on the important column

Descriptive Statistics					
QTY		Discount		Total Revenue	
Mean	1.158940397	Mean	29215.90909	Mean	4562947.348
Standard Error	0.04416776689	Standard Error	8334.028577	Standard Error	26533.36884
Median	1	Median	0	Median	4275000
Mode	1	Mode	0	Mode	7200000
Standard Deviat	0.5427426061	Standard Deviat	110563.3831	Standard Deviat	2672109.169
Sample Variance	0.2945695364	Sample Variance	12224261688	Sample Variance	7140167411078
Kurtosis	32.3989341	Kurtosis	24.36321292	Kurtosis	-0.5004116482
Skewness	5.169320621	Skewness	4.699159295	Skewness	0.3731006965
Range	4	Range	800000	Range	13895000
Minimum	1	Minimum	0	Minimum	45000
Maximum	5	Maximum	800000	Maximum	13940000
Sum	175	Sum	5142000	Sum	46277412000
Count	151	Count	176	Count	10142
Largest(1)	5	Largest(1)	800000	Largest(1)	13940000
Smallest(1)	1	Smallest(1)	0	Smallest(1)	45000
Confidence Leve	0.08727132458	Confidence Leve	16448.14232	Confidence Leve	52010.65349
Q1	1	Q1	0	Q1	6980000
Q3	1	Q3	0	Q3	2241000

DescStat | 1 Outliers | StatMeasure | Data Dictionary

Discount

- Uneven Discounting: The Median (0), Mode (0), Q1 (0), and Q3 (0) values indicate that at least 75% of transactions do not involve any discounts at all.
- Extreme Discount Variation: Although the majority do not receive any discounts, the Standard Deviation (Rp110,564) is significantly greater than the Mean (Rp29,216). The Maximum (Rp800,000) indicates the presence of very large discounts but they are rarely given.
- High Skewness (4.70): The discount distribution is highly skewed to the right, dominated by 0 values, with a long "tail" of large discounts.

3. Descriptive Analytics

B. Interpret Statistical measurement on the important column

Descriptive Statistics					
QTY		Discount		Total Revenue	
Mean	1.158940397	Mean	29215.90909	Mean	4562947.348
Standard Error	0.04416776689	Standard Error	8334.028577	Standard Error	26533.36884
Median	1	Median	0	Median	4275000
Mode	1	Mode	0	Mode	7200000
Standard Deviat	0.5427426061	Standard Deviat	110563.3831	Standard Deviat	2672109.169
Sample Variance	0.2945695364	Sample Variance	12224261688	Sample Variance	7140167411078
Kurtosis	32.3989341	Kurtosis	24.36321292	Kurtosis	-0.5004116482
Skewness	5.169320621	Skewness	4.699159295	Skewness	0.3731006965
Range	4	Range	800000	Range	13895000
Minimum	1	Minimum	0	Minimum	45000
Maximum	5	Maximum	800000	Maximum	13940000
Sum	175	Sum	5142000	Sum	46277412000
Count	151	Count	176	Count	10142
Largest(1)	5	Largest(1)	800000	Largest(1)	13940000
Smallest(1)	1	Smallest(1)	0	Smallest(1)	45000
Confidence Leve	0.08727132458	Confidence Leve	16448.14232	Confidence Leve	52010.65349
Q1	1	Q1	0	Q1	6980000
Q3	1	Q3	0	Q3	2241000

DescStat | 1 Outliers | StatMeasure | Data Dictionary

Total Reveneu:

- Average Transaction Value: The Mean (Rp4,562,947) and Median (Rp4,275,000) are quite close together. This is the result of a successful data cleaning process (as indicated by the low skewness).
- 75% Transaction Limit: Q3 (Rp2,241,000) indicates that 75% of transactions have a revenue value below Rp2,241,000.
- Variation: Although outliers have been removed, the Standard Deviation (Rp2,672,109) is still very large compared to the Mean. This confirms that transaction values above Q3 (the top 25% of transactions) vary significantly and account for a large portion of the total cumulative revenue (Sum: Rp46.2 million).
- Low Skewness (0.37): This value indicates that the data is relatively normal and not significantly skewed. This makes the Mean a reliable metric for describing typical customer transaction values.

Key Business Insight

Purchasing patterns indicate customer **reluctance to purchase** more than one item in a single transaction. More than half of TokoBli's **transactions** are for a **single product**.

TokoBli's current discount policy **doesn't reach** the majority of customers. The fact that 75% of transactions (as of Q3) did not receive a discount **suggests** that incentives are **only applied** to the top 25% of transactions or to **very specific** and **exclusive promotional schemes**.

Although Revenue data is now relatively normally distributed and the average transaction appears high (around IDR 4.5 million), **three-quarters of all transactions (75%)** are valued **below IDR 2,241,000.** This means that the high mean value is heavily driven by the top 25% of transactions, which are very large in value (exceeding IDR 2.2 million).

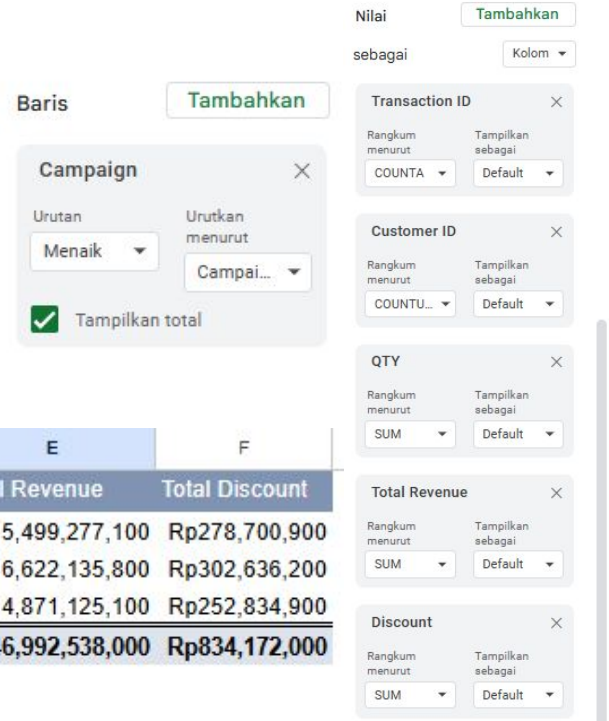
MILESTONE 2 - TokoBli Campaign Evaluation & Product Page Analysis

4. Exploratory Data Analysis (EDA)

A. Compare total transactions, customers, products sold, total revenue, and total discount campaign budget across the three campaigns. Present summary tables and charts, then discuss insights.

To compare the desired values, we need to select all from the sheets dataset cleaning > insert > pivot table > insert on new sheets (Rename to EDA) > Create.

For rows, we have to drag the campaign column to get a comparison of each value for the date each campaign was held. Then, drag the required columns to the values in the pivot table with different summary types. Here, COUNTA is used for Transaction values, and COUNTUNIQUE is used for Customer values.

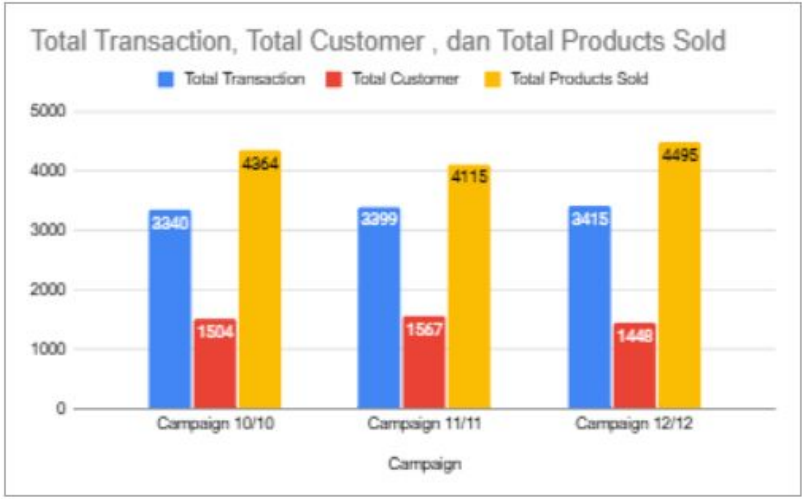


The results of the pivot table. It's better to rename the columns to make it easier to create a chart.

	A	B	C	D	E	F
1	Campaign	Total Transactior	Total Customer	Total Products S	Total Revenue	Total Discount
2	Campaign 10/10	3340	1504	4364	Rp15,499,277,100	Rp278,700,900
3	Campaign 11/11	3399	1567	4115	Rp16,622,135,800	Rp302,636,200
4	Campaign 12/12	3415	1448	4495	Rp14,871,125,100	Rp252,834,900
5	**Grand Total**	**10154**	**4282**	**12974**	**Rp46,992,538,000**	**Rp834,172,000**

4. Exploratory Data Analysis (EDA)

A. Compare total transactions, customers, products sold, total revenue, and total discount campaign budget across the three campaigns. Present summary tables and charts, then discuss insights.



Campaign 12/12 had the **most transactions** (3,415) and the **most products sold** (4,495). However, Campaign 11/11 had the **highest number of customers** (1,567). This shows that Campaign 11/11 **attracted** the most customers, but Campaign 12/12 was the **most effective** at generating transactions and sales volume from a smaller number of customers.

Aggressive discount strategies on 11/11 have proven **effective** in **increasing revenue** per customer and per transaction, **even though** the number of transactions and products sold are **not as high as on 12/12**. This indicates that purchases on 11/11 tend to be for **higher-priced products** or in **larger shopping baskets.**

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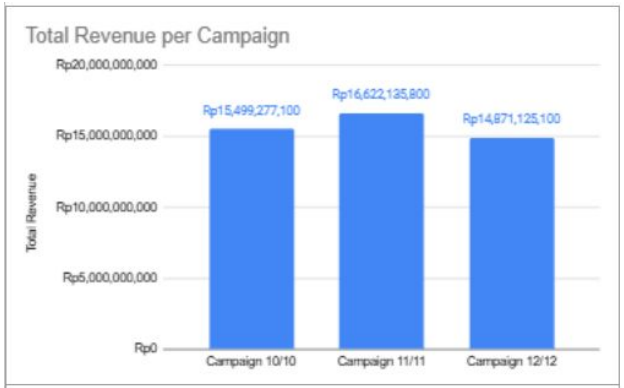

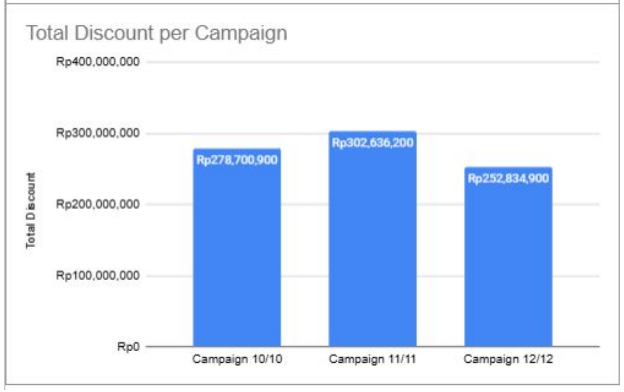

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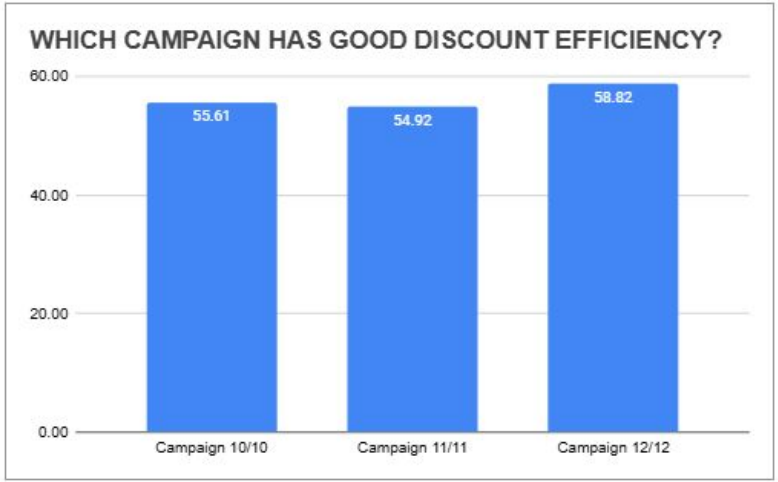
4. Exploratory Data Analysis (EDA)

B. Differences in ratio of revenue vs. campaign discount budget by campaign period; show calculations, charts, and insights.

Campaign	Total Revenue	Total Discount	Revenue/Discount Ratio
Campaign 10/10	Rp15,499,277,100	Rp278,700,900	55.61
Campaign 11/11	Rp16,622,135,800	Rp302,636,200	54.92
Campaign 12/12	Rp14,871,125,100	Rp252,834,900	58.82
Grand Total	**Rp46,992,538,000**	**Rp834,172,000**	56.33

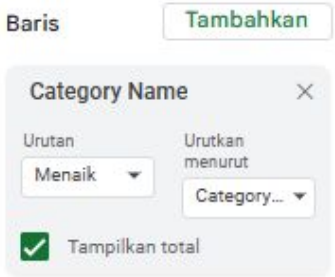


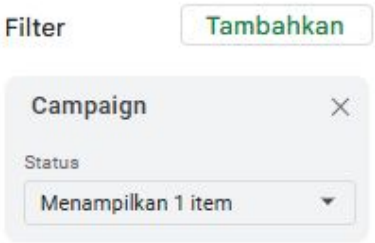
Create a new summary table using the pivot table method as before, but only in the form of total revenue and total discount. Select the column after the total discount column with the column name Revenue/Discount Ratio > type the formula =Revenue Cell / Discount Cell. Select range of columns Campaign and Revenue/Discount Ratio > Click chart.

12/12 had **the highest** revenue-to-discount ratio (58.8), making it the **most efficient campaign**. Every Rp1 discount generated Rp58.8 in revenue. This indicates that 12/12 was **able to drive sales volume without relying on large discounts**. 11/11, while generating the highest revenue, had **the lowest efficiency ratio** (54.9). This means this campaign **required larger discounts** to generate revenue, making the discount's contribution to revenue **less effective** than other campaigns.

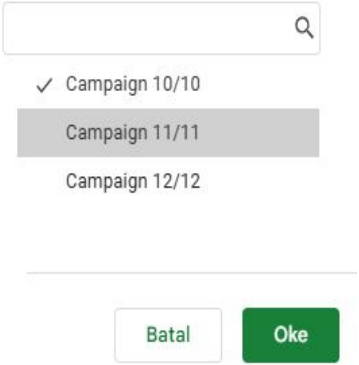
4. Exploratory Data Analysis (EDA)

C. Best-performing product category for transactions, customers, units sold, revenue, and low campaign budget—breakdown per campaign. Present findings and business meaning.





Copy paste the pivot table twice then **select a different campaign** in the filter, so that you have **3 pivot tables for each campaign**



Create a new summary table using the pivot table method as before. For rows, we have to drag the **category name column** to get Best-performing product category per campaign and drag the **campaign column** to **filter** on pivot table.

Category Name	Total Transaction	Total Customer	Total Products Sold	Total Revenue	Total Discount
Appliances	84	79	87	Rp556,784,000	Rp17,366,000
Beauty & Grooming	630	322	926	Rp3,200,065,500	Rp70,117,500
Books	3	2	3	Rp5,970,000	Rp0
Computing	54	46	62	Rp285,063,000	Rp6,417,000
Entertainment	10	10	20	Rp128,532,000	Rp1,548,000
Health & Sports	288	143	401	Rp1,082,422,000	Rp19,658,000
Home & Living	283	184	311	Rp1,151,061,000	Rp17,219,000
Kids & Baby	160	101	182	Rp563,593,000	Rp8,397,000
Men's Fashion	614	430	641	Rp3,371,160,000	Rp33,760,000
Mobiles & Tablets	274	204	310	Rp1,084,970,000	Rp16,440,000
Others	2	2	3	Rp12,500,000	Rp0
School & Education	53	32	57	Rp172,862,000	Rp1,648,000
Soghaat	325	131	414	Rp1,117,046,600	Rp22,007,400
Superstore	328	120	708	Rp1,358,118,000	Rp45,652,000
Women's Fashion	232	159	239	Rp1,409,130,000	Rp18,471,000
Grand Total	**3340**	**1504**	**4364**	**Rp15,499,277,100**	**Rp278,700,900**

4. Exploratory Data Analysis (EDA)

C. Best-performing product category for transactions, customers, units sold, revenue, and low campaign budget—breakdown per campaign. Present findings and business meaning.

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Appliances	84	79	87	Rp556,784,000	Rp17,366,000
Beauty & Grooming	630	322	926	Rp3,200,065,500	Rp70,117,500
Books	3	2	3	Rp5,970,000	Rp0
Computing	54	46	62	Rp285,063,000	Rp6,417,000
Entertainment	10	10	20	Rp128,532,000	Rp1,548,000
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Women's Fashion	232	159	239	Rp1,409,130,000	Rp18,471,000
Grand Total	**3340**	**1504**	**4364**	**Rp15,499,277,100**	**Rp278,700,900**

Campaign 10/10Men's Fashion has the most transactions (614), the most customers (430), the most products sold (641), the highest revenue (Rp. 3,371,160,000) and Books, Others has the lowest discount (Rp. 0).

Category Name	Total Transaction	Total Customer	Total Products Sold	Total Revenue	Total Discount
Appliances	106	94	139	Rp717,723,000	Rp15,497,000
Beauty & Grooming	434	254	541	Rp1,883,670,500	Rp32,145,500
Books	16	12	16	Rp85,901,000	Rp619,000
Computing	63	57	68	Rp340,769,000	Rp6,681,000
Entertainment	27	26	28	Rp140,527,000	Rp2,133,000
Health & Sports	612	135	935	Rp2,050,742,000	Rp23,458,000
Home & Living	297	189	329	Rp1,339,742,000	Rp22,655,000
Kids & Baby	103	70	112	Rp417,368,000	Rp9,202,000
Men's Fashion	544	361	592	Rp3,076,678,000	Rp33,092,000
Mobiles & Tablets	253	194	270	Rp997,191,000	Rp14,139,000
School & Education	35	25	39	Rp95,602,000	Rp2,843,000
Soghaat	218	98	292	Rp628,350,400	Rp14,659,600
Superstore	384	177	796	Rp1,280,959,200	Rp45,762,800
Women's Fashion	323	190	338	Rp1,815,902,000	Rp29,948,000
Grand Total	**3415**	**1448**	**4495**	**Rp14,871,125,100**	**Rp252,834,900**

Category Name	Total Transaction	Total Customer	Total Products Sold	Total Revenue	Total Discount
Appliances	105	84	118	Rp701,922,000	Rp28,898,000
Beauty & Grooming	239	126	285	Rp1,010,663,800	Rp24,408,200
Books	3	3	3	Rp14,106,000	Rp424,000
Computing	38	27	41	Rp157,050,000	Rp4,010,000
Entertainment	13	11	13	Rp72,540,000	Rp1,830,000
Health & Sports	294	82	409	Rp1,074,092,000	Rp22,498,000
Home & Living	235	160	261	Rp980,455,000	Rp27,745,000
Kids & Baby	149	79	160	Rp388,691,000	Rp20,599,000
Men's Fashion	1347	877	1431	Rp8,246,126,000	Rp59,184,000
Mobiles & Tablets	160	123	175	Rp637,921,000	Rp14,269,000
School & Education	11	8	11	Rp33,202,000	Rp588,000
Soghaat	93	56	115	Rp362,828,000	Rp7,442,000
Superstore	560	139	939	Rp2,040,230,000	Rp60,780,000
Women's Fashion	152	95	154	Rp902,309,000	Rp29,961,000
Grand Total	**3399**	**1567**	**4115**	**Rp16,622,135,800**	**Rp302,636,200**

Campaign 11/11

Men's Fashion has the most transactions (1,347), the most customers (877), the most products sold (1,431), the highest revenue (Rp8,246,126,000), and Books has the lowest discount (Rp424,000).

Campaign 12/12

Health & Sports has the most transactions (612) and has the most products sold (935), Men's Fashion has the most customers (361) and has the highest revenue (Rp3,076,678,000), and Books has the lowest discount (Rp619,000).

4. Exploratory Data Analysis (EDA)

C. Best-performing product category for transactions, customers, units sold, revenue, and low campaign budget—breakdown per campaign. Present findings and business meaning.

Campaign	Best Transactions	Best Customers	Best Units Sold	Best Revenue	Lowest Discount
10/10	Men's Fashion	Men's Fashion	Men's Fashion	Men's Fashion	Books / Others
11/11	Men's Fashion	Men's Fashion	Men's Fashion	Men's Fashion	Books
12/12	Health & Sports	Men's Fashion	Health & Sports	Men's Fashion	Books

The Beauty, Health, Home, and Mobile categories are **quite strong**, but still **lag far behind Fashion**. 10/10 is most **effective for Fashion** due to high interest and volume. Categories with low discounts, such as Books, can be maintained without significant discounts due to their low elasticity.

Men's Fashion revenue exceeded IDR 8.2 billion, indicating that the 11/11 event is **effective in the mid-to-high price category**. The 11/11 event is well-suited for categories **sensitive to large promotions**, especially Fashion. The large discount budget on 11/11 makes this campaign large, but it is **more wasteful in terms of efficiency** than 12/12.

Health & Sports took over the top spot for transactions & products. Superstores **showed high engagement** despite the increased discount costs. The 12/12 campaign **attracted shoppers in practical and household** needs categories like Health & Sports. **Smaller discounts but still high volume indicate better efficiency.**

Men's Fashion **dominated** and **won across almost all metrics and campaigns**, especially revenue. It consistently remains a key driver of e-commerce revenue.

Health & Sports was the category with the **highest transactions** and the **most products sold on December 12**, demonstrating significant potential for the year-end period. It deserves to be highlighted during a mega campaign.

Books continued to sell even **without discounts**, demonstrating demand that is independent of campaigns, as **users buy out of necessity.**

4. Exploratory Data Analysis (EDA)

D. Product category with highest ratio of revenue to campaign discount budget—show which category, calculations, and insights.

Campaign 10/10

Category Name	Total Revenue	Discount	Revenue/Discount Ratio
Appliances	Rp556,784,000	Rp17,366,000	32.06
Beauty & Groom	Rp3,200,065,500	Rp70,117,500	45.64
Books	Rp5,970,000	Rp0	#DIV/0!
Computing	Rp285,063,000	Rp6,417,000	44.42
Entertainment	Rp128,532,000	Rp1,548,000	83.03
Health & Sports	Rp1,082,422,000	Rp19,658,000	55.06
Home & Living	Rp1,151,061,000	Rp17,219,000	66.85
Kids & Baby	Rp563,593,000	Rp8,397,000	67.12
Men's Fashion	Rp3,371,160,000	Rp33,760,000	99.86
Mobiles & Tablet	Rp1,084,970,000	Rp16,440,000	66.00
Others	Rp12,500,000	Rp0	#DIV/0!
School & Educat	Rp172,862,000	Rp1,648,000	104.89
Soghaat	Rp1,117,046,600	Rp22,007,400	50.76
Superstore	Rp1,358,118,000	Rp45,652,000	29.75
Women's Fashic	Rp1,409,130,000	Rp18,471,000	76.29
Grand Total	**Rp15,499,277,100**	**Rp278,700,900**	55.61

Campaign 11/11

Category Name	Total Revenue	Discount	Revenue/Discount Ratio
Appliances	Rp701,922,000	Rp28,898,000	24.29
Beauty & Groom	Rp1,010,663,800	Rp24,408,200	41.41
Books	Rp14,106,000	Rp424,000	33.27
Computing	Rp157,050,000	Rp4,010,000	39.16
Entertainment	Rp72,540,000	Rp1,830,000	39.64
Health & Sports	Rp1,074,092,000	Rp22,498,000	47.74
Home & Living	Rp980,455,000	Rp27,745,000	35.34
Kids & Baby	Rp388,691,000	Rp20,599,000	18.87
Men's Fashion	Rp8,246,126,000	Rp59,184,000	139.33
Mobiles & Tablet	Rp637,921,000	Rp14,269,000	44.71
School & Educat	Rp33,202,000	Rp588,000	56.47
Soghaat	Rp362,828,000	Rp7,442,000	48.75
Superstore	Rp2,040,230,000	Rp60,780,000	33.57
Women's Fashic	Rp902,309,000	Rp29,961,000	30.12
Grand Total	**Rp16,622,135,800**	**Rp302,636,200**	54.92

Campaign 12/12

Category Name	Total Revenue	Discount	Revenue/Discount Ratio
Appliances	Rp717,723,000	Rp15,497,000	46.31
Beauty & Groom	Rp1,883,670,500	Rp32,145,500	58.60
Books	Rp85,901,000	Rp619,000	138.77
Computing	Rp340,769,000	Rp6,681,000	51.01
Entertainment	Rp140,527,000	Rp2,133,000	65.88
Health & Sports	Rp2,050,742,000	Rp23,458,000	87.42
Home & Living	Rp1,339,742,000	Rp22,655,000	59.14
Kids & Baby	Rp417,368,000	Rp9,202,000	45.36
Men's Fashion	Rp3,076,678,000	Rp33,092,000	92.97
Mobiles & Tablet	Rp997,191,000	Rp14,139,000	70.53
School & Educat	Rp95,602,000	Rp2,843,000	33.63
Soghaat	Rp628,350,400	Rp14,659,600	42.86
Superstore	Rp1,280,959,200	Rp45,762,800	27.99
Women's Fashic	Rp1,815,902,000	Rp29,948,000	60.64
Grand Total	**Rp14,871,125,100**	**Rp252,834,900**	58.82

Create a new summary table using the pivot table method as before, but only in the form of total revenue and total discount. Select the column after the total discount column with the column name Revenue/Discount Ratio > type the formula =Revenue Cell / Discount Cell.

4. Exploratory Data Analysis (EDA)

E. Producing overall insights and business recommendations

Campaign	Kategori dengan Ratio Tertinggi	Alasan
10/10	School & Education (~105)	Diskon kecil tapi revenue tetap tinggi.
11/11	Men's Fashion (~139)	Promo sangat efisien, volume besar.
12/12	Books (~138.7)	Revenue tinggi meski diskon yang sangat kecil.

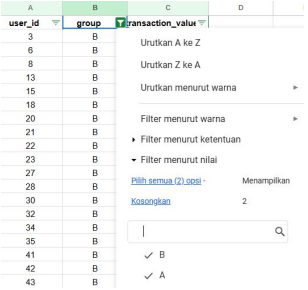
- **High-ratio categories** = highly efficient discount usage. They don't need large discounts to increase revenue.
- Categories like Men's Fashion, Books, and School & Education **can focus on SMALL, HIGH-RETURN PROMOTIONS**.
- Low-ratio categories (Kids, Appliances, Superstore, etc.) **require relatively large discounts to drive revenue**. Suitable strategies include bundle promotions, free shipping, and cashback (not cash discounts).
- **Big Trend Campaigns (11/11 & 12/12)**: Fashion and Books consistently perform well. Users are already waiting for big campaigns to shop for fashion. **Discounts don't need to be huge**; they're actually buying because of the event. **Promotions can be shifted from niche categories** (Kids, Supermarket) to efficient categories. **Use psychological triggers**, not big discounts: Countdown timers, limited-time flash deals, "Today only, 11/11, limited stock."
- **Opportunities for Optimization**

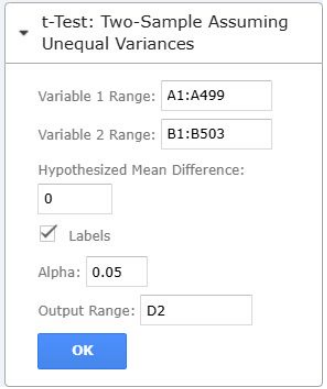
1. Shift discount budgets from low-ROI categories to high-volume but efficient categories.
2. Develop Strategic Bundling
3. Use Margin-Based Discounts: High-margin categories (Fashion, Beauty, Books) can offer small discounts. Low-margin categories (Superstores, Appliances) focus on cashback.

5. Hypothesis Testing

A. Is there any significant difference between Group A and Group B **Transaction Value**?

Hypothesis Statement: Transaction Value between Group A and Group B having significant difference





t-Test: Two-Sample Assuming Unequal Variances

	Group_A	*Group_B*
Mean	746102.6104	830460.1594
Variance	22436486774	33088798170
Observations	498	502
Hypothesized Me	0	
df	965	
t Stat	-8.008050103	
P(T<=t) one-tail	0	
t Critical one-tail	1.646434128	
P(T<=t) two-tail	0	
t Critical two-tail	1.962425286	

Based on P Value < H0 = accepted.
This means the difference is statistically significant.

H0 : Group A TV = Group B TV "The Group A and Group B TV is equal"

H1 : Group A TV != Group B TV "The Group A and Group B TV not equal"

α : 0.05 "The Level of Significance (Alpha) is 5%"

P-Value (two-tail) : 0 "P-Value < alpha, therefore the H1 is accepted"

Conclusion : **H1 condition is accepted**, which means Transaction Value from Group A and Group B is not equal or **having significant difference.**

Group B (new design) generated much higher transaction value

Make new sheets name Hypothesis > select header of columns at sheets dataset > select filter menu > select one of Group A > copy all rows from transaction_value > paste in sheets hypothesis A1 > do the same step, but select filter of Group B.

Do step like page 14 on this slide, but choose t-Test: Two-Sample Assuming Unequal Variances for the XLminer. > select data range of Group A and B > click input Var 1 and 2 range box > fill 0 on Hypothesized Mean Difference > checklist on labels > select new column for result > click output box > OK.

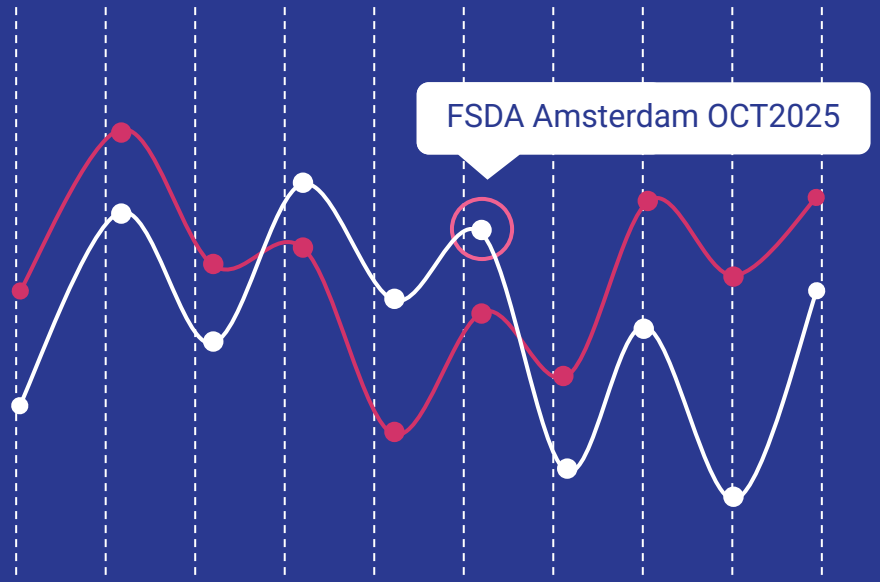
5. Hypothesis Testing

B. Producing business recommendations **(OBPIR Framework)**

Based on the A/B testing results, the recommendation is to **roll out the new design to all users**. The primary objective of this test is to i**ncrease transaction value per user during the promotional period.** After two weeks of testing, the new design in **Group B showed a significant increase compared to the old design**, with average transactions increasing from Rp746,102 to Rp830,460. The t-test results showed a p-value of 0, indicating that this difference is statistically proven and not a random occurrence. The key insight is that the **new design significantly increased transaction value** through a more effective user experience that drives purchases. Therefore, the new design **should be rolled out gradually to minimize risks**, with performance monitoring on metrics such as transactions, bounce rate, and conversion. **Risks to consider** include potential long-term changes in user behavior, differences in performance on mobile devices, and **potential technical load** on the page. However, the significant benefits demonstrated indicate that full implementation of the new design is the right step to increase TokoBli's revenue.

Conclusion

A/B testing results show that the **new product detail page design** significantly increased transaction value, with **average spending by users in Group B** significantly higher than the old design. This **provides strong evidence** that improvements to the display and purchase flow can drive **more effective** purchasing decisions to **maximize revenue in future campaigns**. TokoBli can **capitalize on this insight** by **fully implementing the new design**. Furthermore, TokoBli can **combine** this new design with **campaign strategies** such as personalized product recommendations, flash deals, and bundling on product pages, which now have the potential to have a **greater impact** because the **purchase funnel is more optimized**. By integrating **A/B testing findings** into campaign execution, TokoBli can significantly increase conversion rates, average order values, and total revenue in the next promotional period.

Terima Kasih

Indira Destriana Anjani